

Digital Humanities and Chinese Studies

Instructor: Xuezhao Li

Course Description:

This entry-level graduate course introduces the core concepts, tools, and workflows of Digital Humanities as well as their applications in Chinese Studies. It begins with discussions on what Digital Humanities is, its potential and limitations, and an overview of how DH methods have been used to study Chinese literature, history, and culture. Then it moves to hands-on engagement with key DH methods, including social network analysis, GIS mapping, stylometry, topic modeling, and other computational text analysis techniques. It also covers basic Python programming and teaches students to use coding assistants like Claude Code (tool usage aligns with the university's AI policy) to enhance their coding practices. Students work with large collections of texts (Chinese poems and novels), digital archives (historical maps), and bibliographical data to apply these methods in practice. High proficiency in Chinese is not required, as students may also work with English translations where appropriate.

This course adopts a lab-style format, with practicum sessions following theoretical instruction. Students will be guided through the entire workflow of a DH project, from data collection and preprocessing to analysis and visualization. They will learn how to gather data from major Chinese bibliographic databases, historical map archives, and full-text corpora. They will develop skills in data cleaning and processing for English and literary/vernacular Chinese. In addition, they will participate in collaborative projects that make use of digital platforms such as Gephi, ArcGIS, MARKUS, and Voyant. Besides learning key DH methods, students build the capacity to navigate uncertainty in practice, adapt to an exploratory learning environment, and critically reflect on the affordances of DH approaches in Chinese Studies.

Course Goals:

By the end of this course, students should be able to:

1. Define distant reading, text mining, and data as the field uses those terms, and evaluate a published DH study, naming at least two methodological limitations of its approach.
2. Locate and retrieve material from at least three major Chinese Studies resources (CBDB, historical map archives, a full-text corpus), and state what questions each source can and cannot answer.
3. Execute a complete workflow — gather, clean, analyze, visualize — for at least three of the five methods taught, using Gephi, ArcGIS, Voyant, or Python as the method requires.
4. Design and complete an independent or collaborative DH project, present its findings in writing and visualization, and account for the decisions made at each stage of the workflow.
5. Diagnose at least two ways Chinese text breaks a standard processing pipeline — word segmentation, variant characters, classical versus vernacular register — and apply a documented workaround.

Assessment:

1. Reading journals: Students will keep a weekly journal of assigned readings, which records their observations and critical reflections.
2. After each practicum session, students will present a brief project showcase and submit a short lab report documenting their engagement with Digital Humanities methods and practices. It can include questions: what research questions have they explored? what skills have they learned? Have their research questions been completely solved? Are there any remaining problems, and what future directions can the project take?
3. Students will complete a DH project, either independently or collaboratively. They will present their findings about this project and produce a substantial research essay.

Weighting and Alignment:

Component	Weight
Reading journals (weekly)	15%
Lab reports and project showcases (5 × 7%)	35%
Final project — presentation	15%
Final project — research essay and visualization	35%

Alignment: Reading journals assess Goal 1. Lab reports and showcases assess Goals 2, 3 and 5. The final project assesses Goals 3, 4 and 5.

Final Project Rubric:

Each dimension is scored Exceeds, Meets, or Developing. A project must reach Meets on Data and on Critical reflection to pass, whatever the other scores.

Dimension	Exceeds	Meets	Developing
Research question	Answerable with the data at hand, and could not have been answered by reading alone; scope matched to one term's work.	Clear and answerable, though scope is somewhat wide or narrow for the data.	A topic rather than a question, or not answerable by the method chosen.
Data	Sources named with provenance; cleaning decisions documented with their reasoning; a reader could reproduce the dataset.	Sources named and cleaning described; a reader could approximately reconstruct the dataset.	Sources vague or cleaning undocumented; the dataset cannot be reconstructed.
Method	Justified against at least one alternative considered and rejected; parameters reported.	Suits the question and is executed correctly; parameters reported.	Applied by default, or executed with errors that affect the result.

Dimension	Exceeds	Meets	Developing
Interpretation and visualization	Visualization chosen for what the data has to show; the reading distinguishes what the analysis demonstrates from what it merely suggests.	Visualization legible and correctly labelled; interpretation follows from the output.	Visualization decorative or mislabelled; interpretation overstates what the output supports.
Critical reflection	Names what the method could not see and what a reader should therefore not conclude; identifies how Chinese-language material shaped the pipeline.	Names the main limitation of the approach and at least one unresolved problem.	Limitations acknowledged only in general terms, or omitted.

Collaborative projects carry one additional requirement: a half-page statement of who did what, submitted by each member. The score is shared except where these statements disagree.

Weekly Schedule (two classes each week)

Week One:

Class 1: Introduction to the course

In-class discussions of the syllabus, including course goals and assessment methods, and an overview of expectations for each session.

Class 2: Introduction to Digital Humanities

The session introduces key DH concepts—such as close and distant reading, data, and text mining—as well as major DH methods, including social network analysis, GIS mapping, stylometry, topic modeling, and other computational text analysis techniques. It will also provide an overview of the current state of the field and examine the types of research tasks to which these methods have been applied.

Readings:

Underwood, Ted. *Distant Horizons: Digital Evidence and Literary Change*. Chicago: University of Chicago Press, 2019. Preface and Chapter 1, pp. ix-33.

Moretti, Franco. *Distant Reading*. London: Verso, 2013. Chapter 2, pp. 43-62.

Week Two:

Class 1: Critical Digital Humanities

The session discusses the ongoing debates and key concerns in the field of Digital Humanities, as well as the strengths and limitations of DH methods. It also introduces the principles of Critical Digital Humanities and guides students in making critical reflections on DH methodologies.

Readings:

Gold, Matthew K., and Lauren F. Klein. "Introduction." In *Debates in the Digital Humanities 2023*, 1–14. Minneapolis: University of Minnesota Press, 2023.

Dobson, James E. *Critical Digital Humanities: The Search for a Methodology*. Champaign: University of Illinois Press, 2019. Chapter 1, pp. 1-31.

Class 2: The Application of DH Methods in Chinese Studies

The session provides an overview of the recent trends in East Asian, specifically Chinese, digital humanities studies, as demonstrated in both English- and Chinese-language scholarship.

Readings:

Vierthaler, Paul. 2020. "Digital Humanities and East Asian Studies in 2020." *History Compass* 18, no. 11: e12628.

Xiaoguang Wang, Xu Tan, and Huinan Li. "The Evolution of Digital Humanities in China." *Library Trends* 69, no. 1 (2020): 7–29.

Week Three:

Class 1: Introduction to Social Network Analysis

The session introduces the concepts, tools (Gephi, Tableau, NetworkX in Python), and workflows of Social Network Analysis. It also discusses the potential of the network approach to transform historical studies.

Readings:

Ahnert, Sebastian E., et al. *The Network Turn: Changing Perspectives in the Humanities*. Cambridge: Cambridge University Press, 2020.

Bastian, M. et al., "Gephi: An Open Source Software for Exploring and Manipulating Networks." International AAAI Conference on Weblogs and Social Media.

Class 2: Social Network Analysis in Chinese Studies

It introduces major biographical and relational databases in Chinese Studies: China Biographical Database Project (CBDB), Modern China Biographical Database (MCBD), and Taiwan Biographical Ontology (TBIO), etc. It also discusses prior social network analysis projects in Chinese Studies that students can learn from.

Readings:

Chen, Song, and Hongsu Wang. "China Biographical Database (CBDB): A Relational Database for Prosopographical Research of Pre-Modern China." *Journal of Open Humanities Data* 8, (2022): 1-4.

Mazanec, Thomas J. "Networks of Exchange Poetry in Late Medieval China: Notes Toward a Dynamic History of Tang Literature." *Journal of Chinese Literature and Culture* 5, no. 2 (2018): 322–359.

Week Four:

Class 1: Social Network Analysis in Practice

The practicum session guides students through the workflow of SNA projects, from data gathering and processing to data analysis and visualization. Students will gather data from China Biographical Database (CBDB), import it into Gephi, and experiment with the network visuals.

Class 2: Social Network Analysis Project Showcase

Students will present a small SNA project independently or in teams.

Week Five:

Class 1: Introduction to GIS Mapping

The session introduces the concepts, tools (ArcGIS, QGIS, Google Maps), and workflows of GIS Mapping. It also discusses how GIS methods can contribute to a new development of historiography.

Readings:

Gregory, Ian N., and Alistair Geddes. "Introduction." In *Toward Spatial Humanities: Historical GIS and Spatial History*, ix-xxii. Bloomington: Indiana University Press, 2014.

McHaffie, Patrick, Sungsoo Hwang, and Cassie Follett. *GIS: An Introduction to Mapping Technologies*. 2nd ed. Boca Raton, FL: CRC Press, 2023. Chapter 11, "GIS and Digital Humanities," pp. 249-280.

Class 2: GIS Mapping in Chinese Studies

The session introduces major GIS resources in Chinese Studies, including databases such as The China Historical Geographic Information System (CHGIS), and online historical map collections or platforms, such as The WebGIS Platform of Historical Maps of China, Qing Maps, Taiwan Historical Maps, etc. It also discusses prior GIS projects in Chinese Studies that students can learn from.

Readings:

Stanley-Baker, Michael. "Mapping the *Bencao*: Laoguanshan at the Far End of the Yellow River Corridor." *Asian Medicine* 18, no. 1–2 (2024): 260–303.

Zhaopeng, Wang, Qiao Junjun, "Geographic Distribution and Change in Tang Poetry: Data Analysis from the "Chronological Map of Tang-Song Literature," trans. Thomas J. Mazanec, *Journal of Chinese Literature and Culture* 5, no. 2 (2018), 360-374.

Week Six:

Class 1: GIS Mapping in Practice

The practicum session guides students through the workflow of GIS mapping, from data gathering and processing to data analysis and visualization. Students will learn to use the China Historical Geographic Information System (CHGIS) to create customized digital maps. They will also practice producing their own georeferenced maps using ArcGIS, starting from gathering

spatial data and selecting appropriate base maps, and then combining the two into a coherent, layered digital map for analysis and visualization.

Class 2: GIS Project Showcase

Students will present a small GIS project independently or in teams.

Week Seven:

Class 1: Basic Text Analysis with Voyant

In this session, students use Voyant tools to learn basic text analysis skills such as word frequency counting, keyword extraction, word trends, and visualizations like word clouds.

Class 2: Basic Text Analysis in Python

The session introduces basic text analysis techniques in Python, such as word frequency analysis, keyword extraction, collocation analysis, concordance generation, and simple data visualization methods. It introduces Python text analysis packages for both English and Chinese texts.

Week Eight:

Class 1: Computational Text Analysis of Chinese novels or poems

In this practicum session, students will analyze texts of Chinese novels or poems through coding in Python. We will walk through the whole workflow from data gathering, data cleaning, to data analysis and visualization. They can gather the full-text data from the Chinese Text Project or other databases, or by OCR (Optical Character Recognition). Major OCR tools for Chinese, such as Baimiao will be introduced. Data preprocessing steps for Chinese texts, including stopword removal, Punctuation and Special Character Removal, and tokenization using tools such as HanLP. They perform various techniques of text analysis to explore different features of their selected texts. In the process, they can use coding assistants like Gemini Code Assist.

Class 2: Basic Text Analysis Showcase

Students will present a small text analysis project independently or in teams.

Week Nine:

Class 1: Introduction to Topic Modeling

The session introduces a major computational technique of thematic analysis: topic modeling, including its key concepts, tools, and workflows. It compares two popular topic modeling tools: LDA (Latent Dirichlet Allocation) and BERTopic. The core concepts and steps for these two methods will be explained. Questions to discuss also include: what is a topic in topic modeling, and how is it different from the conventional understanding of a literary topic? What are the appropriate models and workflows for your research projects?

Readings:

Mohr, John W., and Petko Bogdanov. "Introduction—Topic Models: What They Are and Why They Matter." *Poetics* 41, no. 6 (2013): 545–569.

Egger, Roman, and Joanne Yu. "A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts." *Frontiers in Sociology* 7 (2022): 1–16.

Class 2: Topic Modeling in Chinese Studies

The session discusses prior topic modeling cases in Chinese Studies that students can learn from.

Readings:

Broadwell, Peter, Jack W. Chen, and David Shepard, "Reading the Quan Tang Shi: Literary History, Topic Modeling, Divergence Measures," *Digital Humanities Quarterly* 13, no. 4 (2019), 1-26.

Ryan Nichols, Edward Slingerland, Kristoffer Nielbo, Uffe Bergeton, Carson Logan, and Scott Kleinman, "Modeling the Contested Relationship between Analects, Mencius, and Xunzi: Preliminary Evidence from a Machine-Learning Approach," *The Journal of Asian Studies* 77, no. 1 (2018), 19–57.

Zhou, Mengyuan. "Three Faces of Heroism: An Empirical Study of Indirect Literary Translation Between Chinese-English-Portuguese of Wuxia Fiction." *Corpus-based Studies across Humanities* 2, no. 1 (2024): 157–186.

Week Ten:

Class 1: Topic Modeling in Practice

In this practicum session, students will make use of the previously preprocessed texts of Chinese novels or poems to do topic modeling analysis. They can either use LDA topic modeling or BERTopic.

Class 2: Topic Modeling Project Showcase

Students will present a small topic modeling project independently or in teams.

Week Eleven: Introduction to Stylometry

Class 1: The session introduces a major computational technique of stylistic analysis: stylometry, including its key concepts, tools, and workflows. It discusses the wide range of cases in which stylometric analysis can be applied: authorship attribution, genre classification, translation style comparison, and detecting stylistic trends. It thus emphasizes the importance of formulating creative and rigorous research questions when designing stylometric analyses. It also introduces text clustering visualization techniques including HCA (Hierarchical Cluster Analysis), PCA (Principal Component Analysis), and t-SNE (t-distributed Stochastic Neighbor Embedding), etc. Students will also reflect on the concept of "style" and critically examine how stylometric approaches contribute to new understandings of literary styles.

Readings:

Rybicki, Jan, Maciej Eder, and David L. Hoover. "Computational Stylistics and Text Analysis", in *Doing Digital Humanities: Practice, Training, Research*. Constance Crompton, Richard J. Lane, and Raymond Siemens (eds), 123–144. London: Routledge, 2016.

Herrmann, Berenike J., Karina van Dalen-Oskam, and Christof Schöch. "Revisiting Style, a Key Concept in Literary Studies." *Journal of Literary Theory* 9, no. 1 (2015): 25–52.

Class 2: Stylometry in Chinese Studies

This session discusses prior stylometric analysis cases in Chinese Studies that students can learn from.

Readings:

Vierthaler, Paul. "Fiction and History: Polarity and Stylistic Gradience in Late Imperial Chinese Literature." *Journal of Cultural Analytics* 1, no. 1 (2016): 1–32.

Kurzynski, Maciej. "The Stylometry of Maoism: Quantifying the Language of Mao Zedong." In *Proceedings of the Joint 3rd International Conference on Natural Language Processing for Digital Humanities and 8th International Workshop on Computational Linguistics for Uralic Languages*, 76–81. Tokyo: Association for Computational Linguistics, 2023.

Haoran Zhu, Lei Lei, and Hugh Craig, "Prose, Verse and Authorship in *Dream of the Red Chamber*: A Stylometric Analysis," *Journal of Quantitative Linguistics* 28, no. 4 (2020): 289–305.

Week Twelve:

Class 1: Stylometry in Practice

In this practicum session, students will make use of the previously preprocessed texts of Chinese novels or poems to do stylometric analysis in Python.

Class 2: Stylometry Project Showcase

Students will present a small stylometry project independently or in teams.

Week Thirteen:

Class 1: Lab Sessions for Final Projects, Part I

In this session, students will work on their own final projects. The instructor will answer their questions and offer help when needed.

Class 2: Lab Sessions for Final Projects, Part II

In this session, students will continue to work on their own final projects.

Week Fourteen:

Class 1: Final Project Presentations, Part I

In this session, students present their final projects (completed or in progress) to the class, introducing their research questions, corpus, methods, findings, and critical questions. They can do it independently or work in teams.

Class 2: Final Project Presentations, Part II

In this session, students continue to present their final projects.